

FAST – NUCES



Object Oriented Programming Lab Manual

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Lab 6 – OOP Functions

Objectives

- Utility functions
 - Accessor functions
 - Mutator functions
-



Read carefully and attempt following task

“TownsvilleGuardian” Class

Create a class named TownsvilleGuardian that represents a hero protecting Townsville from villains like Mojo Jojo.

Private Data Members

The class must contain:

- string heroName → Name of the guardian
- int energyLevel → Energy level (0–100)
- int villainsDefeated → Total villains defeated
- string superPower → Special ability

Constructors

Default Constructor

Initialize to:

- heroName = "New Recruit"
- energyLevel = 60
- villainsDefeated = 0
- superPower = "Chemical X Burst"

Overloaded Constructor

Accept:

- heroName
- energyLevel
- superPower

Set villainsDefeated = 0 by default.

Destructor

Print:

Guardian signing off... Townsville is safe!

1. Accessor Functions (Getters)

Must be const

Must NOT modify data

Required:

- string getHeroName() const
- int getEnergyLevel() const
- int getVillainsDefeated() const

2. Mutator Functions (Setters)

Must modify private data

Must include validation

Required:

setEnergyLevel(int energy)

- Only allow values between 0 and 100

defeatVillain()

- Increase villainsDefeated by 1

3. Utility Functions

These perform actions related to the guardian.

displayStatus()

Print in format:

Hero: Blossom

Energy Level: 85

Villains Defeated: 3

Super Power: Ice Breath

battle(int villainStrength)

If energyLevel > villainStrength:

- Print "Villain defeated!"
- Call defeatVillain()
- Reduce energyLevel by 10

Else:

- Print "Retreat! Call Professor Utonium!"

recharge()

Increase energyLevel by 15 (max 100).

Object Passing & Returning

compareEnergy(TownsvilleGuardian otherGuardian)

Return:

- 1 → If current object has more energy
- -1 → If less
- 0 → If equal

nextBattleReady()

- Return a NEW TownsvilleGuardian object
- Energy increased by 5
- Do NOT modify original object

Main() Requirements

In main():

1. Create one object using default constructor
2. Create one using overloaded constructor
3. Call:
 - All accessor functions
 - All mutator functions
 - All utility functions
4. Compare two guardians
5. Demonstrate object returned from nextBattleReady()
6. Show destructor call at program end



-----Happy Coding 😊-----



Submission Instructions

- Submit your tasks on Google Classroom within given deadline
- Submit solution file of each task separately for example “LAB01-TASK01.cpp”, “LAB01-TASK02.cpp” and so on
- Also submit screenshots of output of each task for example “LAB01-TASK01.png”, “LAB01-TASK02.png” and so on
- “.cpp” is the actual file where all your code is saved.
- Always submit .cpp files when asked for submission
- Example path of .cpp file is

Desktop\Your_Roll_No\LAB01-TASK01\LAB01-TASK01\LAB01-TASK01.cpp

Copy your files from this path

